

Ternary Relational Neural Networks: Context-Gated Signed Evidence as a Hidden-Unit Primitive

A research prototype and exploratory study

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Abstract. Most conventional neural-network units learn a weight vector and compute a projection of the input. Prototype-based networks instead compare an input representation with learned reference points. This paper introduces a small architectural variation intended to make that comparison itself relational: a **Ternary Relational Unit (TRU)** maintains three learned reference states per hidden unit - a positive evidence prototype, a negative evidence prototype, and a neutral/context prototype. The positive and negative references create signed evidence, while the neutral reference controls how strongly that evidence is expressed. The resulting activation is therefore a product of what the input resembles and whether the unit considers that evidence contextually decisive.

The novelty claim is intentionally limited. Prototype layers and positive/negative prototype neurons already exist; the proposed contribution is the specific use of a third neutral/context reference to gate signed prototype evidence inside the hidden unit. In a targeted literature search, no identical architecture was found in the sources examined, but this is not an exhaustive prior-art search. A small PyTorch benchmark on the scikit-learn digits dataset shows that the prototype is viable but not yet state-of-the-art: TRN reaches 92.22% +/- 0.96% clean accuracy and 87.31% +/- 0.98% under Gaussian test noise, essentially matching the dual-prototype baseline.

Keywords: neural architecture, prototype learning, relational neurons, interpretability, robust representation learning, architectural inductive bias

1. Motivation

A standard hidden neuron answers: *How much does this input point in the direction I learned?* A prototype neuron asks: *How similar is this input to the reference I learned?* TRN asks a third question: *Is the input closer to evidence than counter-evidence, and does the unit consider that comparison contextually decisive?* The goal is to encode these three roles structurally rather than forcing a downstream linear layer to reconstruct them implicitly.

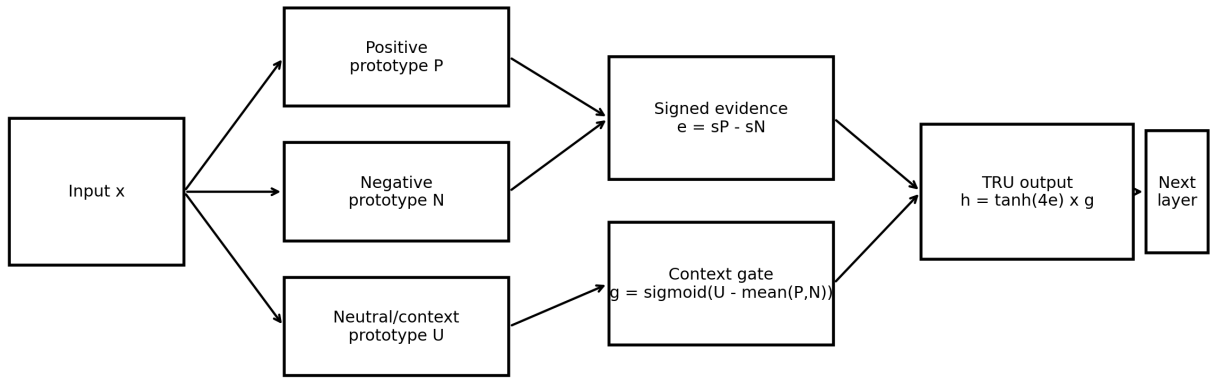
2. Related Work

Prototype-based neural networks are an established family. Deep Case-Based Reasoning networks introduced prototype layers whose units compute distances to learned latent prototypes and use those distances for classification and explanation. Prototype-based neural-network layers have also incorporated vector-quantization ideas. The closest conceptual predecessor found in the targeted search is the +/-ED-WTA work, which models a neuron using a positive prototype and a negative prototype and interprets the neuron through the difference between distances to those references. Prototypical Contrastive Learning uses prototypes for representation learning, but not as the same per-unit relational primitive used here.

3. Proposed Architecture

For hidden unit j , let P_j , N_j , and U_j be the positive, negative, and neutral/context reference vectors. Given x , compute mean-squared distances d_P , d_N , d_U and similarities $s_P = \exp(-d_P/\tau)$, $s_N = \exp(-d_N/\tau)$, $s_U = \exp(-d_U/\tau)$. The signed evidence is $\mathbf{e} = s_P - s_N$. The contextual gate is $\mathbf{g} = \text{sigmoid}(\gamma(s_U - 0.5(s_P + s_N)))$. The final activation is $\mathbf{h} = \tanh(4\mathbf{e}) * \mathbf{g} + \mathbf{b}$, followed by LayerNorm in the implementation.

Ternary Relational Unit (TRU)



The third reference is not another classifier vote; it modulates how confidently the P-vs-N evidence is expressed.

Figure 1. Ternary Relational Unit. Positive and negative references generate signed evidence; the neutral/context reference modulates its amplitude.

4. Intuition

Imagine a unit trained to detect a handwritten 3. P can encode a characteristic 3-like state, N a confusing counter-pattern, and U a region in representation space in which the unit expects this P-vs-N comparison to be meaningful. A strong positive-vs-negative comparison outside the contextual region is suppressed; a similarly strong comparison inside it is allowed through. Thus the unit separates evidence from contextual trust.

5. Implementation

The implementation is written in PyTorch. The full TRN stacks two TRU layers followed by a linear classifier. A layer of width h with input width d stores three prototype matrices of size $h \times d$, plus temperature and context-gain parameters, biases, and normalization parameters. The method is therefore parameter-heavy and computationally more expensive than a standard MLP.

6. Experimental Setup

Dataset: scikit-learn digits, 1,797 examples, 64 input features, 10 classes. An 80/20 stratified train/test split is used. Optimization uses AdamW, learning rate 0.002, weight decay $1e-4$, full-batch training, and 25 epochs. Three seeds (0, 1, 2) are evaluated. The noisy test condition adds Gaussian noise with $\sigma = 0.20$ only at evaluation time and clips pixels to $[0,1]$.

7. Results

Model	Parameters	Clean accuracy	Noisy accuracy
MLP	8,970	71.57% +/- 4.85%	66.39% +/- 6.41%
DualPrototype	17,546	92.13% +/- 1.12%	87.22% +/- 0.28%
TRN (ours)	25,866	92.22% +/- 0.96%	87.31% +/- 0.98%

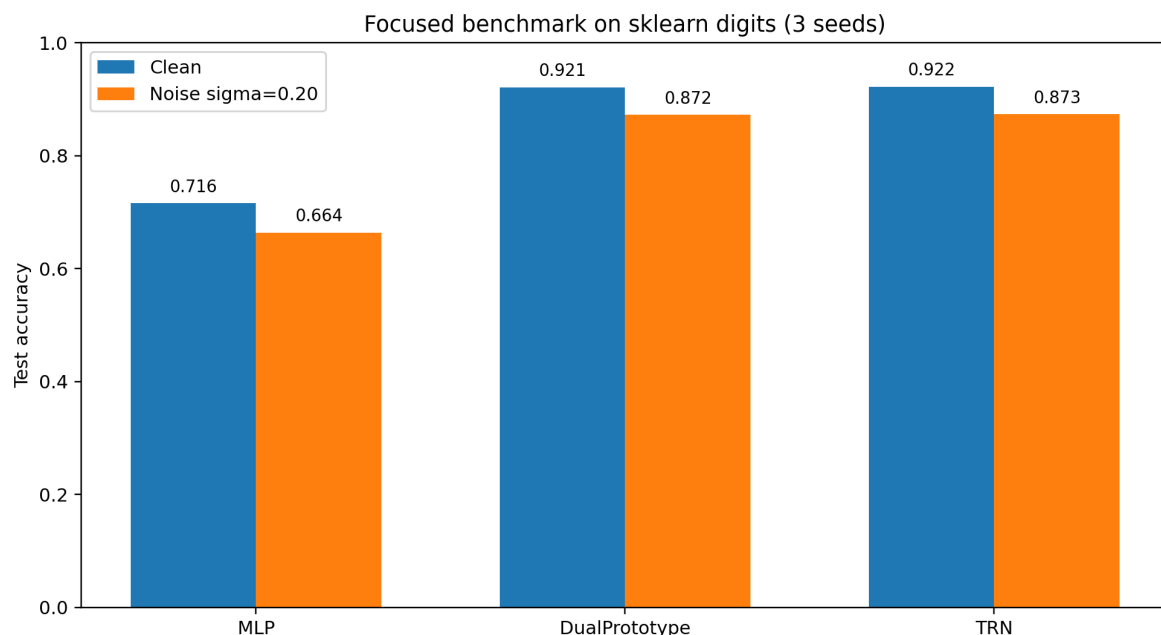


Figure 2. Focused benchmark on the digits dataset. TRN is essentially tied with the dual-prototype baseline on corrupted inputs and shows a small clean-data improvement.

8. Interpretation

The central numerical result is not the 0.09 percentage-point clean-accuracy difference; that is small relative to the uncertainty of a three-seed study. The useful finding is that introducing explicit contextual gating did not collapse the performance of the dual-prototype architecture and produced a distinct mechanism that can be studied independently. At present, the evidence is insufficient to claim general superiority.

9. Limitations and Scientific Status

This is a research prototype. The literature search was targeted, not exhaustive, so the safe novelty statement is that this specific mechanism was not found in the searched sources. The benchmark is small, the TRN has more parameters, and the result needs confirmation on larger datasets and parameter-matched baselines. The current context-gate formula is also partly heuristic.

10. Future Work

The next experiment should equalize parameter counts, then evaluate on MNIST, Fashion-MNIST, CIFAR-10 embeddings, synthetic manifolds, and several corruption families. Additional directions include learnable distance metrics, dynamic context prototypes, convolutional TRN layers, and a probabilistic interpretation in which the neutral prototype represents local uncertainty or evidence reliability.

11. Conclusion

TRN proposes a simple but structurally different hidden unit: evidence is represented by a positive-vs-negative prototype comparison, while a third neutral prototype controls whether that evidence should be expressed. The initial CPU-scale experiment shows that this idea is trainable and competitive with a close prototype baseline on a small benchmark. The result is best viewed as a credible starting point for further research rather than a finished neural architecture.

References

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